

Vijay Raj Venugopal

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Professional Summary

Mechanical Engineer with experience in field engineering, infrastructure construction, robotics systems integration, and facilities operations. Hands-on experience working with mechanical systems, equipment assembly, troubleshooting, and on-site technical support. Skilled in CAD, simulation, ROS2 robotics systems, data analysis, and quality control. Strong ability to bridge design and real-world implementation through practical, hands-on problem solving. Seeking an entry-level Mechanical, Manufacturing, or Systems Engineering role.

Education

Master of Engineering – Mechanical Engineering | Old Dominion University | GPA: 3.54 | Dec 2025
Bachelor of Engineering – Mechanical Engineering | Bir Tikendrajit University | GPA: 3.33 | Dec 2023
Java Full Stack Development Certification | Quest Innovative Solutions Pvt. Ltd. | Apr 2023

Certifications

Six Sigma Green Belt (Lean Six Sigma – Accredited) | Udemy (AIGPE) | April 2026
HVAC Maintenance from Zero to Hero (DUAL HVAC Certification – Certificate of Completion) | Udemy | April 2026
AIGPE™ Certified Failure Mode and Effects Analysis (FMEA) Specialist | Udemy (AIGPE) | April 2026
Java Full Stack Development Certification | Quest Innovative Solutions Pvt. Ltd. | April 2023

Technical Skills

- AutoCAD, Gazebo, RViz2
- Python, Java, MATLAB, Linux (Ubuntu)
- ROS2, SLAM (Cartographer), Navigation2, Path Planning (A*, Dijkstra)
- LiDAR, Camera Systems, Ultrasonic Sensors
- Microsoft Excel, Word, PowerPoint

Professional Experience

Facilities Management Assistant – Old Dominion University (Nov 2024 – Dec 2025)

- Managed 20+ facility work orders weekly across campus buildings, ensuring timely completion and minimal disruption.
- Coordinated with cross-functional maintenance teams to maintain safety standards and operational efficiency.
- Prioritized multiple service requests, improving task turnaround and operational workflow.

Engineer Trainee – STRIA Constructions (Oct 2021 – Nov 2022)

- Monitored gas pipeline construction activities including trenching, joint coating, crossings, lowering, and backfilling, with hands-on involvement in site operations and equipment coordination.
- Ensured compliance with engineering drawings, industry codes, and quality control procedures while supporting on-site inspections and mechanical work execution.
- Gained hands-on experience working with construction equipment, pipeline materials, and field tools, assisting in installation, alignment, and troubleshooting of mechanical components.
- Prepared progress reports and project documentation supporting quality control and client handover.

Site Engineer – SAS Electricals (Jun 2020 – Oct 2021 | Sep 2023 – Dec 2023)

- Supervised installation of municipal water pipelines for infrastructure projects, ensuring alignment with project specifications.
- Coordinated 10–20 field workers, managing daily site activities and task assignments.
- Conducted material inspections, trench depth measurements, and safety compliance checks.

Trainee Software Engineer – Quest Innovative Solutions (Jan 2023 – Oct 2023)

- Assisted in debugging and optimizing Java-based applications, improving system reliability and performance.
- Participated in full-stack application development using Java technologies.
- Collaborated with development teams to test features and resolve technical issues.

Projects

Autonomous Indoor Robot System (ROS2) – Master’s Project

Built an autonomous robot using ROS2, SLAM, LiDAR, and Navigation2 for indoor navigation.

Solar Assisted Heat Pump Water Heating System – Undergraduate Project

Designed a system, recovering waste heat from AC units combined with solar collectors for water heating.

Leadership & Volunteering

Rise Up Forum – Volunteer (2019–2023)

Y’s Men International – Youth Club President (2016–2022)

MRA – Volunteer

Languages

English (Professional), Malayalam (Native), Tamil (Conversational), Hindi (Basic)